假設檢定和平均值的比較

Hypothesis Testing and Comparison of Means

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1 Student's *t*-test

Arguments of `t.test()`

- `alternative`: alternative hypothesis
  - `"two.sided"` ($H_0 : \mu_1 = \mu_2$; $H_1 : \mu_1 \neq \mu_2$)
  - `"greater"` ($H_0 : \mu_1 \leq \mu_2$; $H_1 : \mu_1 > \mu_2$)
  - `"less"` ($H_0 : \mu_1 \geq \mu_2$; $H_1 : \mu_1 < \mu_2$)
- `mu = 0`:
  - for one-sample test, true value of the mean

  - for two-sample test, difference in the two means

- `paired = FALSE`: whether two samples are related (paired)
- `var.equal = FALSE`: whether two population variances are equal
- `conf.level = 0.95`: confidence level

See `?t.test` for more details.

1.1 One-sample *t*-test

Exercise (1)

欲調查某廠牌泡麵防腐劑含量是否超標 (≤ 3 ppm)，隨機抽取該廠牌 10 包泡麵，並測量每包泡麵的防腐劑含量。

```
pres <- read.csv("data/preservative.csv")
str(pres)
```

```
'data.frame':   10 obs. of  1 variable:
 $ ppm: int  3 4 5 3 4 5 4 4 2 5
```

1. 請以「單尾檢定」檢驗此泡麵防腐劑含量是否符合政府所制定的國家標準 (3 ppm)。

   令該廠牌泡麵的防腐劑平均含量為 μ，欲檢定之假說為：

$$\begin{cases} H_0 : \mu \leq 3 \text{ ppm} \\ H_1 : \mu > 3 \text{ ppm} \end{cases}$$

   ```
   t.test(pres$ppm, alternative = "greater", mu = 3)
   ```

   ```
    One Sample t-test

   data:  pres$ppm
   t = 2.862, df = 9, p-value = 0.00936
   alternative hypothesis: true mean is greater than 3
   95 percent confidence interval:
    3.323548      Inf
   sample estimates:
   mean of x
         3.9
   ```

```r
# How to get p-value?
1 - pt(2.862, df = 9)
```

```
[1] 0.009359624
```

Conclusion

此廠牌泡麵防腐劑含量之樣本平均值為 3.9 ppm，在顯著水準 0.05 下

- 95% 單尾信賴區間 $(3.32, \infty)$ 未包含 3
- p-value $= 0.0094$ 小於顯著水準

可以拒絕 H_0。此廠牌泡麵防腐劑平均含量顯著高於政府所制定的國家標準。

2. 請以「雙尾檢定」檢驗此泡麵防腐劑含量是否符合政府所制定的國家標準 (3 ppm)。

   令該廠牌泡麵的防腐劑平均含量為 μ，欲檢定之假說為：

$$\begin{cases} H_0 : \mu = 3 \text{ ppm} \\ H_1 : \mu \neq 3 \text{ ppm} \end{cases}$$

```r
t.test(pres$ppm, alternative = "two.sided", mu = 3)
```

```
 One Sample t-test

data:  pres$ppm
t = 2.862, df = 9, p-value = 0.01872
alternative hypothesis: true mean is not equal to 3
95 percent confidence interval:
 3.188628 4.611372
sample estimates:
mean of x
      3.9
```

```r
# How to get p-value?
(1 - pt(2.862, df = 9)) * 2
```

```
[1] 0.01871925
```

Conclusion

此廠牌泡麵防腐劑含量之樣本平均值為 3.9 ppm，在顯著水準 0.05 下

- 95% 雙尾信賴區間 (3.19, 4.61) 未包含 3
- p-value = 0.0187 小於顯著水準

可以拒絕 H_0。此廠牌泡麵防腐劑平均含量與政府所制定的國家標準有顯著差異。

1.2 Paired two-sample *t*-test

Exercise (2)

欲探討牙周再生手術是否能改善牙周病人口腔狀況，研究者收集 10 位牙周病人接受牙周再生手術前後之牙周囊袋深度 (probing pocket depth)。

```
ppd <- read.csv("data/ppd.csv")
str(ppd)
```

```
'data.frame':   10 obs. of  3 variables:
 $ ppd0: int  10 7 8 7 8 7 8 7 9 8
 $ ppd1: int  3 1 2 2 1 3 3 2 2 3
 $ diff: int  7 6 6 5 7 4 5 5 7 5
```

- `ppd0`: 治療前牙周囊袋深度 (單位: mm)。
- `ppd1`: 治療後牙周囊袋深度 (單位: mm)。
- `diff`: 治療前減治療後的牙周囊袋深度 (單位: mm)。

牙周囊袋深度變化是指同一位病人在治療前後的變化，因此同一病人 2 次牙周囊袋深度具有相關性。

令治療前與治療後牙周囊袋平均深度分別為 μ_1 和 μ_2，欲檢定之假說為：

$$\begin{cases} H_0 : \mu_1 - \mu_2 \leq 0 \\ H_1 : \mu_1 - \mu_2 > 0 \end{cases}$$

進行 paired t-test 可使用兩種語法，結果一致:

```r
# general interface
t.test(ppd$ppd0, ppd$ppd1, alternative = "greater", paired = TRUE)

# formula interface
t.test(Pair(ppd0, ppd1) ~ 1, data = ppd, alternative = "greater")
```

```
    Paired t-test

data:  ppd$ppd0 and ppd$ppd1
t = 17.015, df = 9, p-value = 1.879e-08
alternative hypothesis: true mean difference is greater than 0
95 percent confidence interval:
 5.085915      Inf
sample estimates:
mean difference
            5.7
```

Conclusion

接受牙周再生手術的牙周病人，牙周囊袋深度平均減少 5.7 mm，在顯著水準 0.05 下

- 95% 單尾信賴區間 $(5.09, \infty)$ 未包含 0
- p-value $= 1.88 \times 10^{-8}$ 小於顯著水準

可以拒絕 H_0。接受牙周再生手術能顯著減少牙周囊袋平均深度。

Note

將同一病人治療前後之牙周囊袋深度相減，再進行 one-sample t-test，結果一致。

```r
t.test(ppd$diff, alternative = "greater")
```

```
    One Sample t-test

data:  ppd$diff
t = 17.015, df = 9, p-value = 1.879e-08
alternative hypothesis: true mean is greater than 0
95 percent confidence interval:
 5.085915      Inf
sample estimates:
```

```
mean of x
     5.7
```

1.3 Independent two-sample *t*-test

- 兩母體變異數相同 ($\sigma_1^2 = \sigma_2^2 = \sigma^2$)
  - Sample variances:

$$s_1^2 = \frac{\sum_{i=1}^{n_1}(X_i - \bar{X})^2}{n_1 - 1} \quad \text{and} \quad s_2^2 = \frac{\sum_{j=1}^{n_2}(Y_j - \bar{Y})^2}{n_2 - 1}$$

  - Pooled sample variance:

$$s_p^2 = \frac{\sum_{i=1}^{n_1}(X_i - \bar{X})^2 + \sum_{j=1}^{n_2}(Y_j - \bar{Y})^2}{n_1 + n_2 - 2} = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

  - Test statistic:

$$T = \frac{\bar{X} - \bar{Y}}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{df}, \quad df = n_1 + n_2 - 2$$

- 兩母體變異數不同 ($\sigma_1^2 \neq \sigma_2^2$) — **Welch's *t*-test**
  - Test statistic:

$$T = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim t_{df}, \quad df = \frac{(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2})^2}{\frac{(\frac{s_1^2}{n_1})^2}{n_1 - 1} + \frac{(\frac{s_2^2}{n_2})^2}{n_2 - 1}}$$

Exercise (3)

欲研究懷孕婦女吸菸是否會影響嬰兒體重，隨機抽取 14 名吸菸的孕婦和 15 名無吸菸的孕婦，測量嬰兒出生體重。

```
smk_wt <- readxl::read_xlsx("data/smoking_birth_weight.xlsx")
str(smk_wt)
```

```
tibble [29 x 2] (S3: tbl_df/tbl/data.frame)
 $ smoking    : chr [1:29] "smokers" "smokers" "smokers" "smokers" ...
 $ birthweight: num [1:29] 3.18 2.74 2.9 3.27 3.65 3.42 3.23 2.86 3.6 3.65 ...
```

檢定前先進行基本的探索性分析：

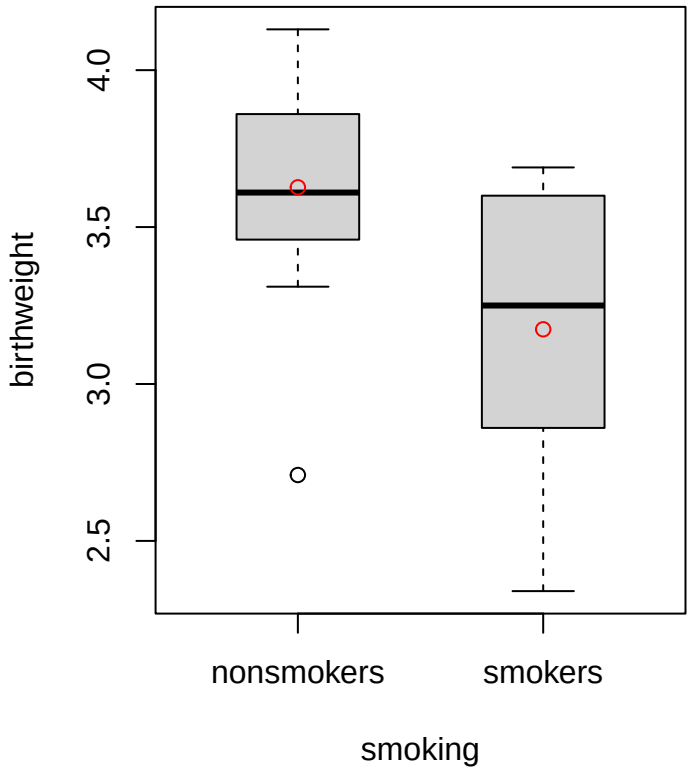
```
table(smk_wt$smoking)
```

```
nonsmokers    smokers
        15          14
```

```
aggregate(birthweight ~ smoking, data = smk_wt,
          FUN = \(x) c(Mean = mean(x), SD = sd(x)))
```

```
     smoking birthweight.Mean birthweight.SD
1 nonsmokers        3.6266667      0.3584025
2    smokers        3.1742857      0.4631450
```

```
boxplot(birthweight ~ smoking, data = smk_wt, boxwex = 0.5)
points(aggregate(birthweight ~ smoking, data = smk_wt, FUN = mean), col = "red")
```



令吸菸與無吸菸孕婦生下的嬰兒平均體重分別為 μ_1 和 μ_2，欲檢定之假說為：

$$\begin{cases} H_0 : \mu_1 = \mu_2 \\ H_1 : \mu_1 \neq \mu_2 \end{cases}$$

How to know $\sigma_1^2 = \sigma_2^2$ or not?

_F_-test of equality of variances (two samples)

$$\begin{cases} H_0 : \sigma_1 = \sigma_2 \\ H_1 : \sigma_1 \neq \sigma_2 \end{cases}$$

```
var.test(birthweight ~ smoking, data = smk_wt)
```

```
    F test to compare two variances

data:  birthweight by smoking
F = 0.59884, num df = 14, denom df = 13, p-value = 0.353
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.1943104 1.8036316
sample estimates:
ratio of variances
         0.5988364
```

變異數同質性檢定之 p-value $= 0.353$，在顯著水準 0.05 下無法拒絕 H_0，表示吸菸與無吸菸孕婦生下的嬰兒體重之變異數無顯著差異。因此，在進行 t-test 時需設定 `var.equal = TRUE`:

```
t.test(birthweight ~ smoking, data = smk_wt, var.equal = TRUE)
```

```
    Two Sample t-test

data:  birthweight by smoking
t = 2.9535, df = 27, p-value = 0.006437
alternative hypothesis: true difference in means between group nonsmokers and group smokers
95 percent confidence interval:
 0.1381077 0.7666542
sample estimates:
mean in group nonsmokers    mean in group smokers
                3.626667                 3.174286
```

Conclusion

吸菸孕婦生下的嬰兒平均體重，比無吸菸孕婦少了 3.63–3.17=0.46 kg，在顯著水準 0.05 下

- 95% 信賴區間 (0.138, 0.767) 未包含 0
- p-value = 0.006437 小於顯著水準

可以拒絕 H_0。吸菸與無吸菸孕婦生下的嬰兒平均體重有顯著差異。

2 Analysis of Variance (ANOVA)

2.1 Prerequisites

1. Independence of observations
2. Normality
   - **Shapiro-Wilk test** — `shapiro.test()`
   - **Q-Q plot** — `qqplot()`
3. Equality of variances (Homoscedasticity)
   - **Bartlett's test** — `bartlett.test()`:
     對資料的常態性較為敏感；資料偏離常態分布時 type I error rate 增加。
   - **Levene's test** — `car::leveneTest()`:
     對資料偏離常態分布較為穩健；然而，當資料確實來自常態分布時，其檢定力 (power) 通常低於 Bartlett's test。

! Important

- 若資料明顯違反常態性假設, 可改用無母數 (non-parametric) 檢定方法
  - **Kruskal-Wallis test** — `kruskal.test()`
- 若資料近似常態分布，但變異數不相等, 可改用
  - **Welch's ANOVA** — `oneway.test(..., var.equal = FALSE)`

2.2 事後檢定 (post-hoc test)

若有 3 組獨立樣本，需進行 $C_2^3 = 3$ 次成對比較。令一次成對比較犯錯的可容許機率為 α^*，整體犯錯的可容許機率為 α，則

$$\begin{aligned}\alpha &= P(3 \text{ 次成對比較至少犯一錯}) \\ &= 1 - P(3 \text{ 次成對比較皆不犯錯}) \\ &= 1 - (1-\alpha^*)^3 \\ &= 3\alpha^* - 3\alpha^{*2} + \alpha^{*3} \approx 3\alpha^*\end{aligned}$$

為了避免型 I 誤差膨脹，可將一次成對比較犯錯的機率 α^* 設為 $\frac{\alpha}{3}$ (等同於將每一次成對比較的 p-value 乘以 3)，此做法即為 Bonferroni correction。

Exercise (4)

欲探討三種類型的鐮刀型紅血球疾病 (sickle cell disease) 患者的血紅素濃度是否有差異，研究者收集 41 位患者 並記錄他們的疾病類型 (Hb SC, Hb S/b, Hb SS) 與血紅素濃度。

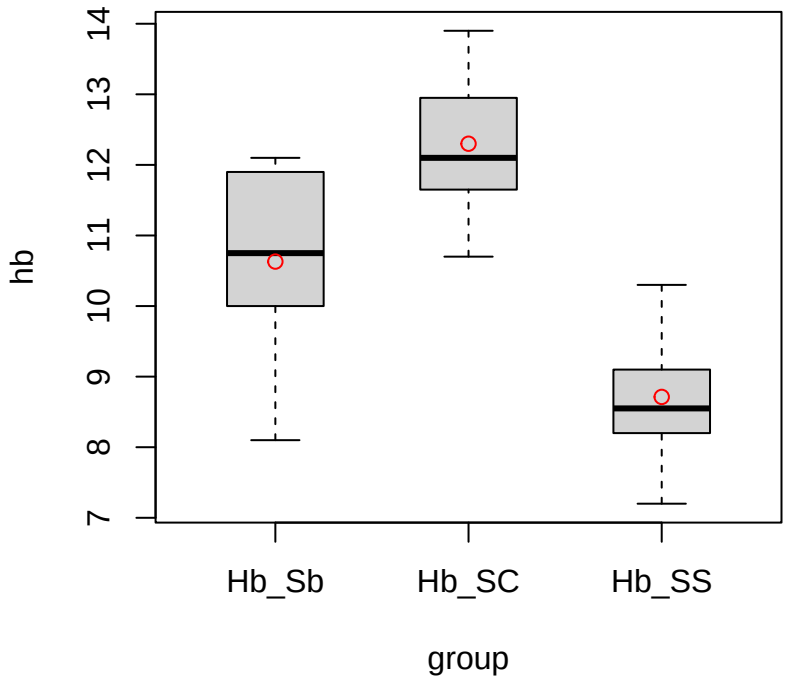
```
hb <- readxl::read_xlsx("data/hemoglobin.xlsx")
str(hb)
```

```
tibble [41 x 2] (S3: tbl_df/tbl/data.frame)
 $ group: chr [1:41] "Hb_SC" "Hb_SC" "Hb_SC" "Hb_SC" ...
 $ hb   : num [1:41] 10.7 11.3 11.5 11.6 11.7 11.8 12 12.1 12.3 12.6 ...
```

```
table(hb$group)
```

```
Hb_Sb Hb_SC Hb_SS
   10    15    16
```

```
boxplot(hb ~ group, data = hb, boxwex = 0.5)
points(aggregate(hb ~ group, data = hb, FUN = mean), col = "red")
```



令三種類型的鐮刀型紅血球疾病患者的血紅素平均濃度分別為 μ_{sb}、μ_{sc}、和 μ_{ss}，欲檢定之假說為：

$$\begin{cases} H_0 : \mu_{sb} = \mu_{sc} = \mu_{ss} \\ H_1 : \text{at least one } \mu_i \neq \mu_j \quad \text{for} \quad i, j = \{sb, sc, ss\} \end{cases}$$

```
hb_aov <- aov(hb ~ group, data = hb)
```

- Test for normality

$$\begin{cases} H_0 : \text{data come from a normal distribution} \\ H_1 : \text{data do not come from a normal distribution} \end{cases}$$

```
shapiro.test(hb_aov$residuals)
```

```
  Shapiro-Wilk normality test

data:  hb_aov$residuals
W = 0.96544, p-value = 0.2427
```

- Test for equality of variances

$$\begin{cases} H_0 : \sigma_{sb}^2 = \sigma_{sc}^2 = \sigma_{ss}^2 \\ H_1 : \text{at least one } \sigma_i^2 \neq \sigma_j^2 \quad \text{for} \quad i, j = \{sb, sc, ss\} \end{cases}$$

```
car::leveneTest(hb ~ group, data = hb)
```

```
Warning in leveneTest.default(y = y, group = group, ...): group coerced to
factor.
```

```
Levene's Test for Homogeneity of Variance (center = median)
      Df F value Pr(>F)
group  2  0.8747 0.4252
      38
```

```
bartlett.test(hb ~ group, data = hb)
```

```
  Bartlett test of homogeneity of variances

data:  hb by group
Bartlett's K-squared = 2.1251, df = 2, p-value = 0.3456
```

- ANOVA

```
summary(hb_aov)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)
group        2  99.89   49.94      50 2.28e-11 ***
Residuals   38  37.96    1.00
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# How to get p-value?
1 - pf(50, df1 = 2, df2 = 38)
```

```
[1] 2.28132e-11
```

Conclusion

p-value $= 2.28 \times 10^{-11}$，在顯著水準 0.05 下可以拒絕 H_0。
三種類型的鐮刀型紅血球疾病患者的血紅素平均濃度並非完全相同。

為了找出哪些類型的鐮刀型紅血球疾病的血紅素平均濃度有差異，需進行事後檢定。

- Bonferroni correction

```
pairwise.t.test(hb$hb, hb$group, p.adjust.method = "bonf")
```

```
  Pairwise comparisons using t tests with pooled SD

data:  hb$hb and hb$group

       Hb_Sb   Hb_SC
Hb_SC 0.00064 -
Hb_SS 8.4e-05 1.1e-11

P value adjustment method: bonferroni
```

這裡的 p-values 已經過調整 (乘以成對比較的次數)，判斷時還是以原本設定的顯著水準 (e.g. 0.05) 當作標準。

Conclusion

在顯著水準 0.05 下，任兩種類型的鐮刀型紅血球疾病的血紅素平均濃度皆有顯著差異。

Note

Bonferroni correction 偏保守 (高估 p-value)，較不容易看出成對比較的差別。可改用其他校正方法:

```
p.adjust.method = "..."
```

- `"holm"`: Holm (1979)
- `"hochberg"`: Hochberg (1988)
- `"hommel"`: Hommel (1988)
- `"BH"` (alias `"fdr"`): Benjamini & Hochberg (1995)
- `"BY"`: Benjamini & Yekutieli (2001)

或是使用 Tukey's HSD (honestly significant difference) test:

```
TukeyHSD(hb_aov)
```

```
  Tukey multiple comparisons of means
    95% family-wise confidence level
```

```
Fit: aov(formula = hb ~ group, data = hb)

$group
                diff        lwr        upr     p adj
Hb_SC-Hb_Sb  1.6700  0.6748973  2.6651027 0.0006147
Hb_SS-Hb_Sb -1.9175 -2.9000852 -0.9349148 0.0000819
Hb_SS-Hb_SC -3.5875 -4.4635296 -2.7114704 0.0000000
```